Solving Bilevel Optimization with Deep Reinforcement Learning

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Introduction

Basic Formulation of Bilevel Optimization:

$$\begin{aligned} \max. \quad & f(x,y) \\ \text{s.t.} \quad & h(x,y) \le 0 \\ & \min \quad g(x,y) \\ & \text{s.t.} \quad p(x,y) \le 0 \end{aligned}$$

1. Two levels of decisions
2. Interactions between levels

Challenge 1. Nested structure:
1) No exact solutions. 2) Approximate & heuristic algorithms:
→ weak optimality, time-consuming.
Challenge 2. Hierarchical decision making with conflicting goals:
→ decisions with long-term impacts.
Challenge 3. Large-scale decisions with dynamic constraints:
→ lower level cannot be linearized (e.g., integer or quadratic).

Bilevel Markov Decision Process (BIMDP)

Agent:

Leader (ULA) for f
- UL state $\boldsymbol{s^u}$: LL variables y
- UL action $\boldsymbol{a^u}$: UL variables x

Follower (LLA) for g
- LL state $\boldsymbol{s^l}$: concatenation of UL and LL variables (x, y)
- LL action $\boldsymbol{a^l}$: LL variables y

Reward:
- objective value
- bonus/penalty term

ULA's Reward Function:

$$\mathcal{R}^u(s^u,a^u,s^{u\prime}) = \begin{cases} f(a^u,s^{u\prime}) + L_p^u, & f(a^u,s^{u\prime}) \ge f^{ub} \\ f(a^u,s^{u\prime}) - L_n^u, & f(a^u,s^{u\prime}) < f^{ub} \end{cases}$$

LLA's Reward Function:

$$\mathcal{R}^l(s^l,a^l,s^{l\prime}) = \begin{cases} -g(a^l,s^{l\prime}) + L_p^l, & g(a^l,s^{l\prime}) \le g^{lb} \\ -g(a^l,s^{l\prime}) - L_n^l, & g(a^l,s^{l\prime}) > g^{lb} \end{cases}$$

Conclusion

Bilevel DRL provides a new perspective of the leader-follower decision making and bilevel programming, which can:

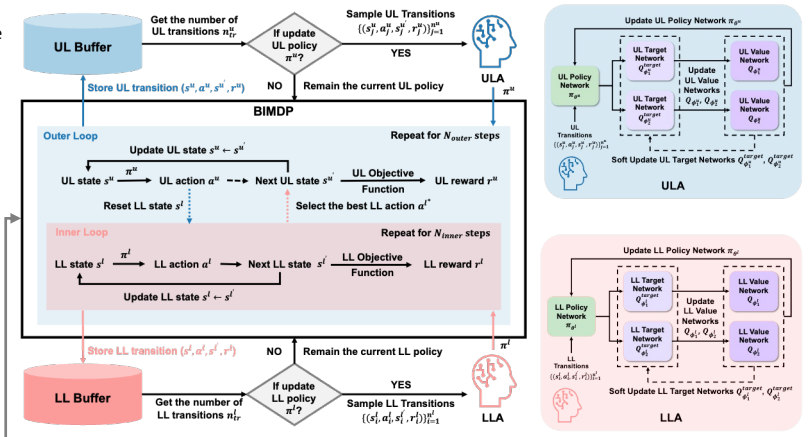
1. effectively solve bilevel optimization models with complex integer constraints,
2. capture the interactive nature between two levels of decision making,
3. efficiently solve large-scale instances compared to state-of-the-art methods.

Reference:

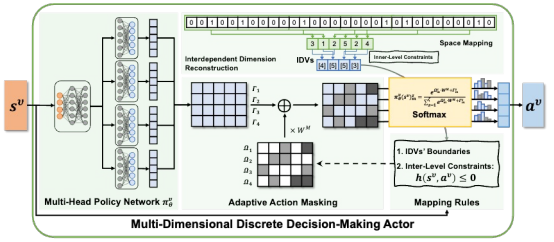
[1]. Xia, X., Ma, Y., Zhang, C., Liu, K. A general deep reinforcement learning framework for bilevel optimization. *Work in progress*.

[2]. Ma, Y., Xia, X., Guo, J., and Zhang, C. (2025). A deep reinforcement learning method solving bilevel optimization for product design considering remanufacturing. *IEEE Transactions on Engineering Management*.

Bilevel Deep Reinforcement Learning (Bilevel DRL)



Policy Network Structure

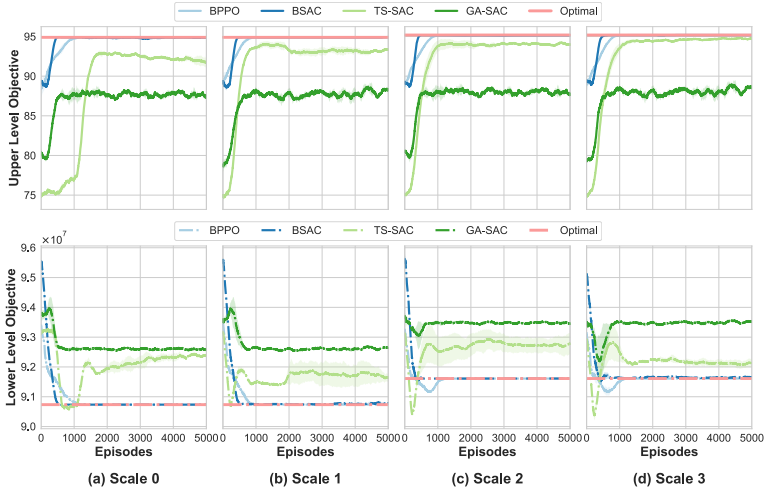


Step 1. Space mapping utilizes 6 dimensions to convey information in the 6×5-dimensional binary format.
Step 2. After interdependent dimension reconstruction, there are 4 IDVs based on inner-level constraints, and each IDV ranges in [4], [5], [5], and [3].
Step 3. Set $\zeta = 5$ and mask 5 for dimension 1 and 4,5 for dimension 4 of the action.

Optimality on Small-Scale Instances

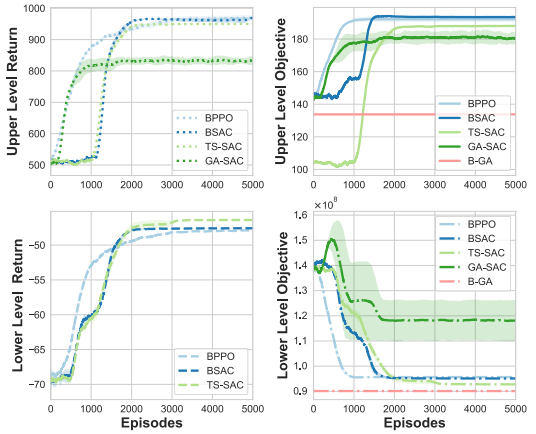
Bilevel DRL:
BSAC: incorporates an off-policy DRL algorithm in a BIMDP with experience buffers
BPPO: incorporates a typical on-policy DRL algorithm in a BIMDP without experience buffers

Baselines:
TS-SAC: a single-level MDP to solve the bilevel optimization without considering interactions
GA-SAC: a bilevel method to solve low-level optimization with a genetic algorithm
Optimal: a conventional bilevel interactive method based on genetic algorithm



Performance on Large-Scale Instances

B-GA: a bilevel interactive method based on genetic algorithm



Running Time Comparison

Method	Running Time (seconds)	Inference Time (seconds)	
		UL Decision	LL Decision
B-GA	32593	325.93	3.26
GA-SAC	82751	3.31	1.65
TS-SAC	16463	0.67	0.21
BPPO	20653	0.83	0.27
BSAC	16713	0.67	0.21